

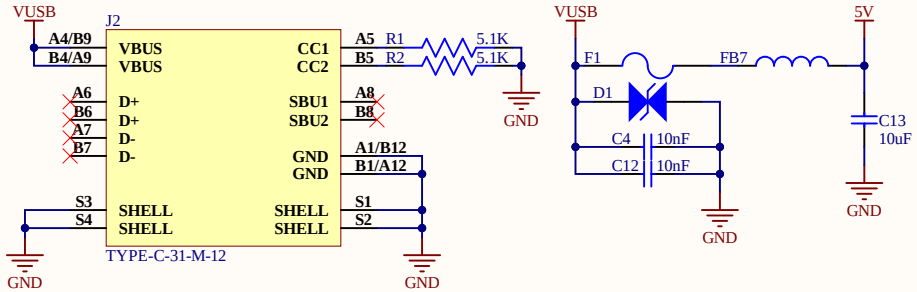
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Top-Sheet.SchDoc	
File: IoT-Speaker.PrjPcb	
Date: 9/6/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 5

VUSB 5V CHARGE INPUT

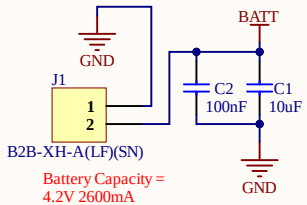
VUSB trace current 2A

5V trace current 2A



Decoupling capacitors should be placed close to each VBUS pin
ESD Diode is to be placed in manner that it has minimum impedance path with VBUS

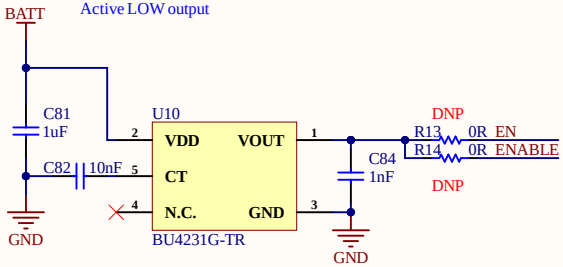
BATTERY INPUT



Battery Capacity = 4.2V 2600mA

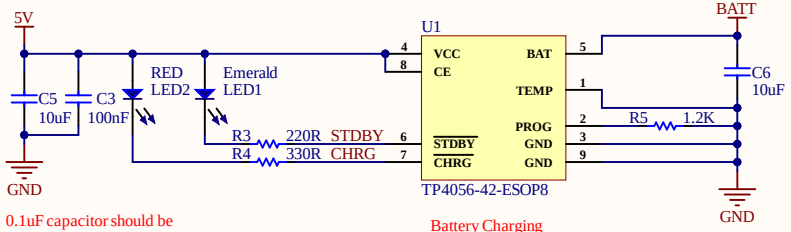
BATTERY SUPERVISOR CIRCUIT

Battery will automatically cutoff below 3.1V
This is open-drain output IC that has Active LOW output



Connect R13 to allow MOSFET control
Connect R14 to allow supervisor circuit control

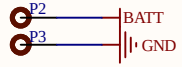
BATTERY CHARGING CIRCUIT

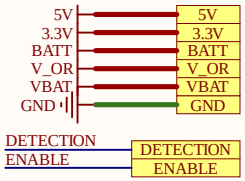


0.1uF capacitor should be placed closest to VCC pin

Battery Charging current = 1000mA

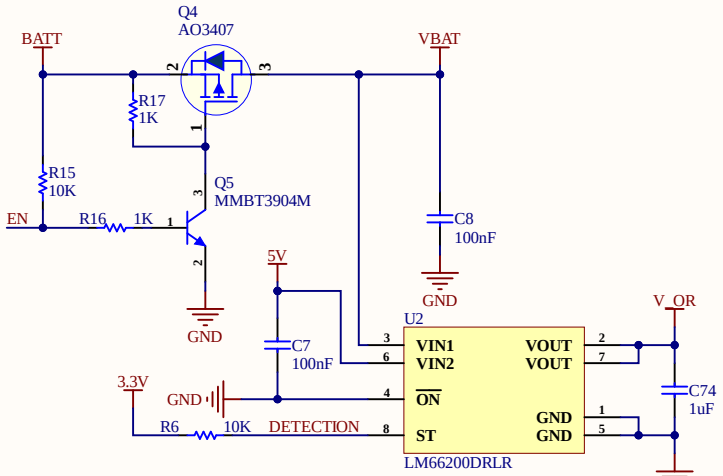
TEST POINTS





POWER SUPPLY ORING CIRCUITRY

The BATT-VBAT circuit is only for testing and maybe removed in production. This is to evaluate if the battery supervisory doesn't quickly activate when charger is connected

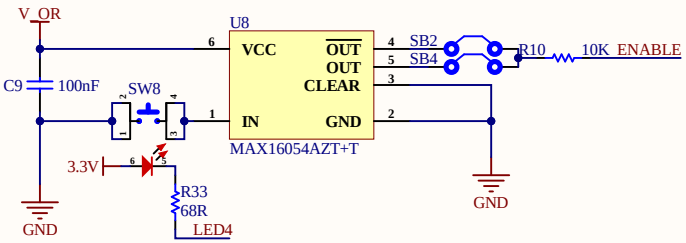


C7 and C8 should be placed as close to their pins as possible
The highest voltage supply is automatically selected. ST pin is used to detect which voltage is currently selected

Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

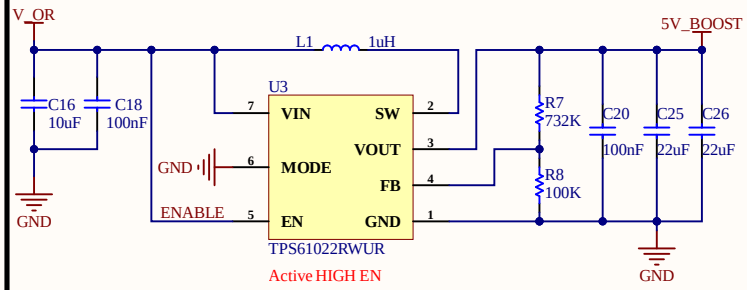
Title: Power-Sheet-1.SchDoc	
File: IoT-Speaker.PrjPcb	
Date: 9/6/2023	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: 5

LATCH ON/OFF CONTROLLER



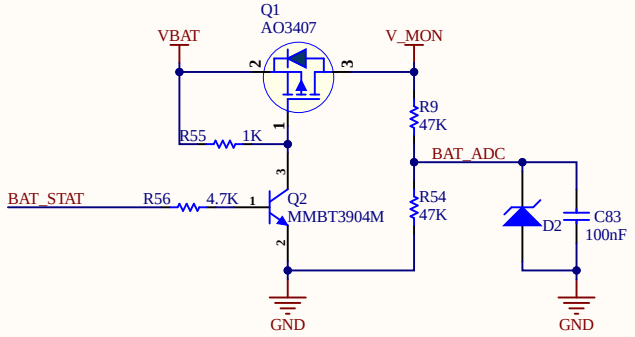
The OUT pin changes state whenever
LOW logic is received at IN pin

BOOST CONVERTER

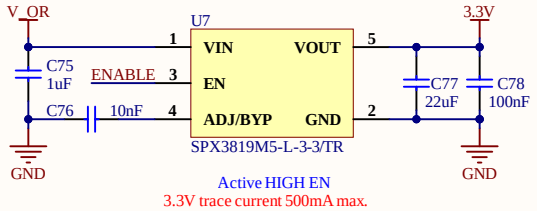


Active HIGH EN

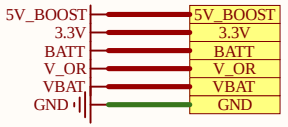
READ BATTERY VOLTAGE



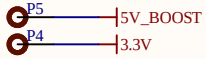
3V3 LDO



Active HIGH EN
3.3V trace current 500mA max.

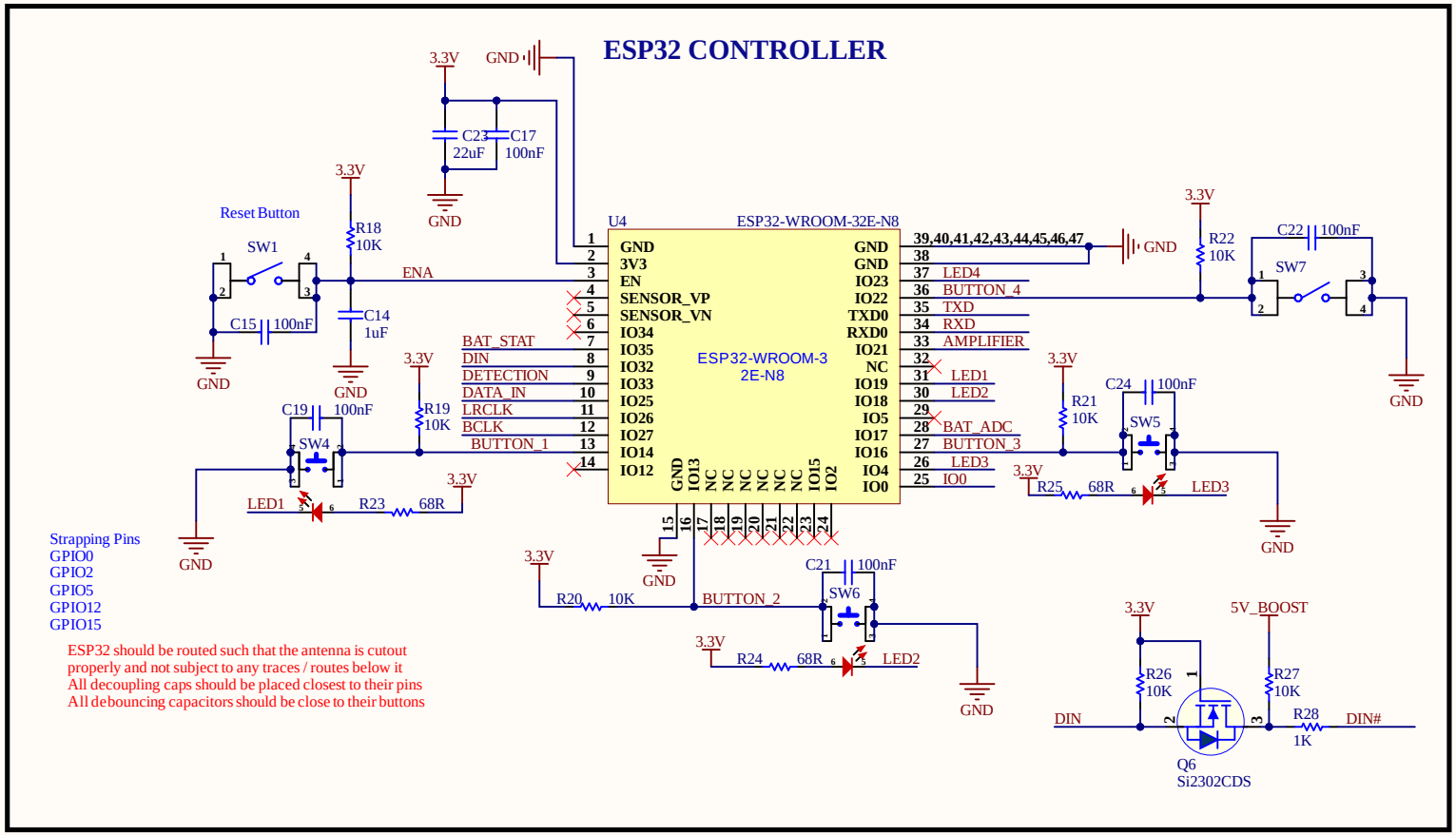


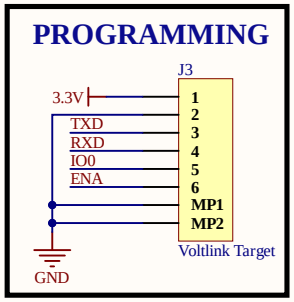
TEST POINTS

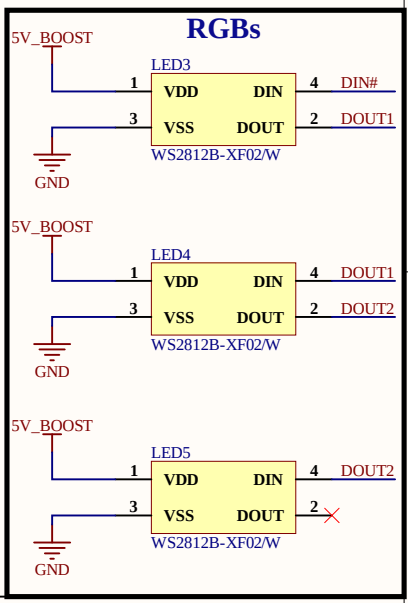


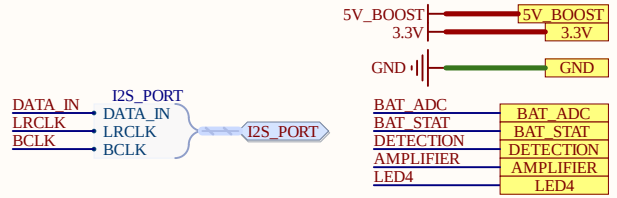
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Power-Sheet-2.SchDoc	
File: IoT-Speaker.PrjPcb	
Date: 9/6/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 5



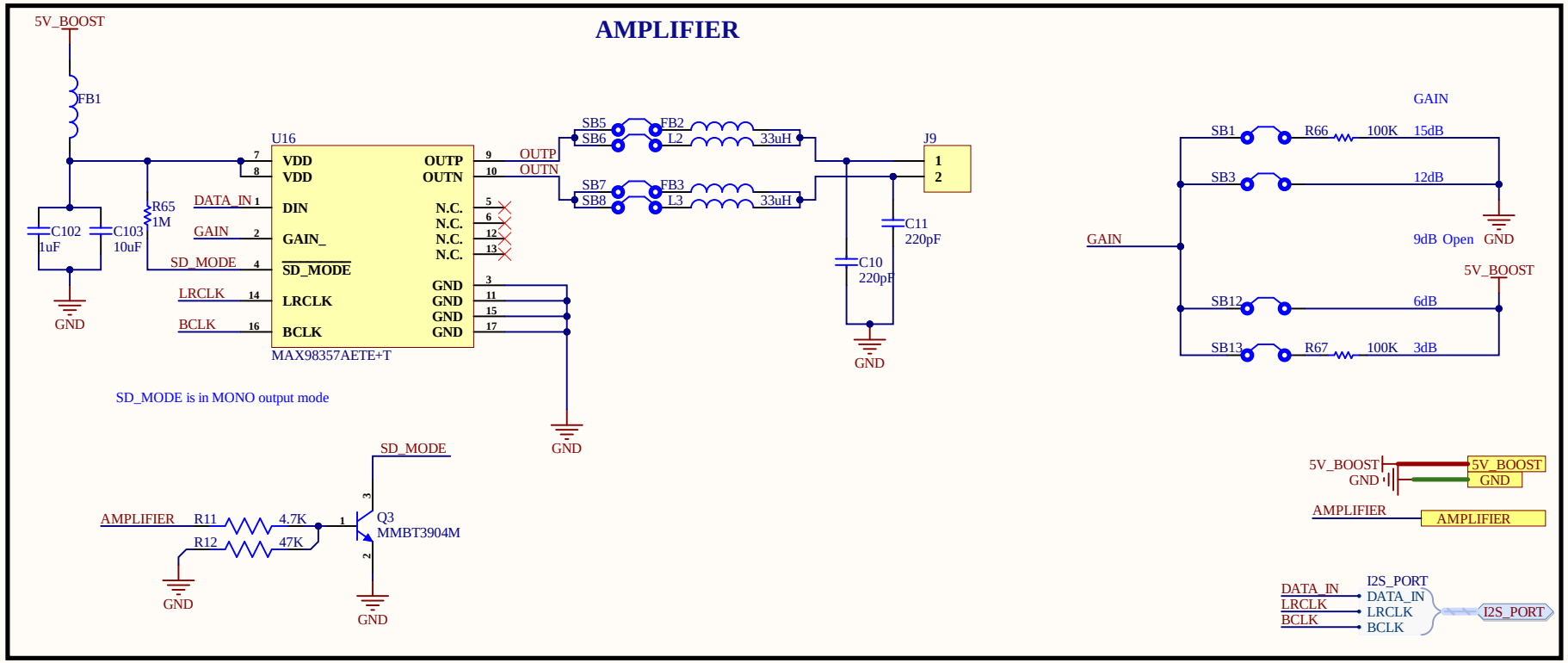






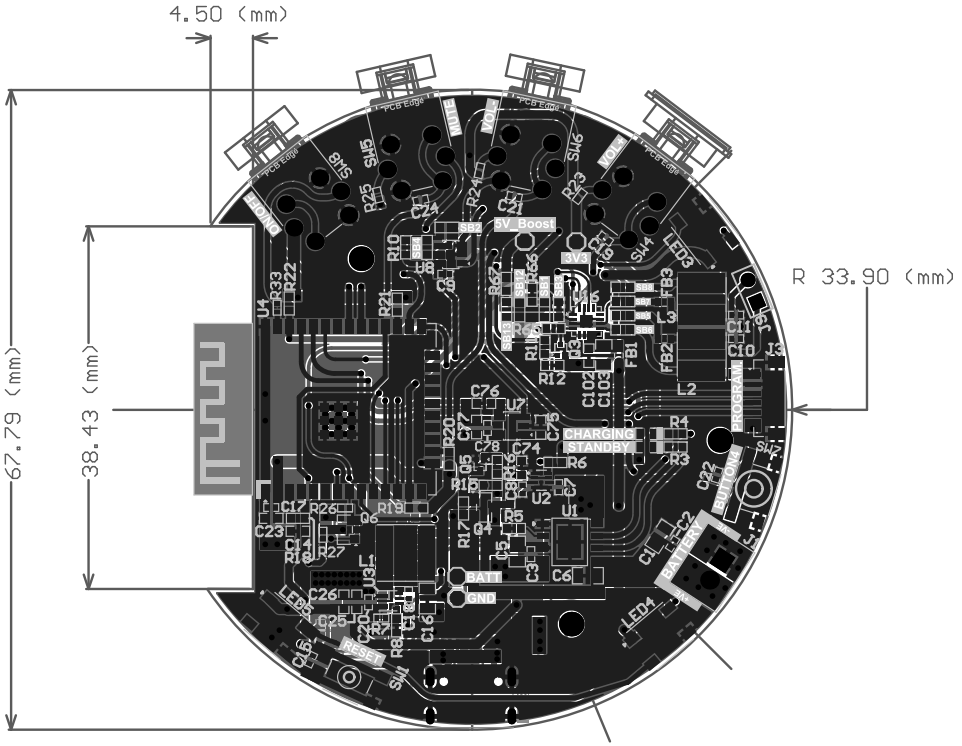
Cannot open file
D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg.
File does not exist.

Title: Controller-Sheet.SchDoc	
File: IoT-Speaker.PrjPcb	
Date: 9/6/2023	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: 5



Cannot open file D:\eSpark-Consultants\eSpark-Hardware\Altium Template\espark-consulting-group-800x300.jpg. File does not exist.

Title: Audio-Amplifier.SchDoc	
File: IoT-Speaker.PrjPcb	
Date: 9/6/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 5



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	3.00mil	3.5	
1	Top Layer		10.00mil		
	Dielectric 1	FR-4	37.00mil	4.8	
2	Bottom Layer		10.00mil		
	Bottom Solder	Solder Resist	3.00mil	3.5	
	Bottom Overlay				
Total board thickness:			63.00mil		